Article

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A corpus-based analysis of object indexing in Hewramî

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Abstract: Hewramî (ISO 639-3) is an Iranic language featuring tense-sensitive alignment, characterised in terms of indexing by employing two sets of bound person markers for expressing direct objects. In TAM categories based on past stem verbs, the direct object is expressed by suffixal person/number morphology. However, the index may be absent under certain conditions. This article explores object indexing in Hewramî, considering its token frequency in discourse and factors conditioning differential object indexing (DOI). The data come from two spoken corpora of over 35,000 words from the Tekht variety of Hewramî. The corpus data were annotated for factors such as animacy, identifiability, person of the referent and textual givenness. The data provide some support for the complementarity hypothesis, meaning that agreement markers slightly favour zero object arguments. The most decisive factor triggering the lack of indexing with overt Os is the co-optation of the agreement slot by a higher-ranked argument. But when taking semantic-referential features into account, the person of the object referent strongly predicts differential object indexing. With the direct object being 3PL and the NP modified, animacy predicts DOI.

Keywords: DOI; person; topicality; animacy; ergativity

1 Introduction

Verbs in some Iranic languages feature tense-sensitive indexing of the O argument, carried out by clitic pronouns in the inflection of present stem verbs, and the verbal person/number suffixes, historically derived from the copula endings, in the inflection of past transitive verbs, see (1). The indexation of O in the past tense via person-number suffixes is a reflex of the ergative construction that started in the late Old Iranic (see Section 3).

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(1) a. *ber-e=**m*** *pişt-û* *bere-w*
take.PRS.IMP-2SG:A=**1SG:O** back-GEN.EZ door.M-GEN.EZ
wiłax-eka
horse.M-DEF.PL.OBL
'Take me behind the door of the horses' stable!' [ŞC.62][1]
(Mohammadirad 2025b) (Hewramî)

b. *mar-êw-î* *gest-**a***
snake-INDF-M.SG.OBL bite.PST-**1SG:O**
'A snake bit me.' [MP.09]
(Mohammadirad in prep) (Hewramî)

The indexing of O via clitic pronouns in present tense constructions has remained pronominal throughout Iranic (Haig 2018), meaning that it could be conditioned to the absence of the co-referential argument in the same local domain. This is equivalent to Haspelmath's (2013) 'pro-index'. In Persian, the object indexing clitic pronouns may double an overt object NP that is definite and thus exhibit incipient traces of obligatory indexing or agreement markers. Still, in indexing O, clitic pronouns have not been grammaticalised as a fully-fledged agreement markers (Rasekh 2014).

On the other hand, there is a significant variation for the indexation of O in past tense constructions (see Section 3 for an overview). This paper focuses on object indexing in past tense constructions of Hewramî, as a case study of differential object indexing within Iranic. By way of example, in both (2a) and (2b), the head noun is modified by a numeral, but only in (2a) does the verb agree with the object NP.

(2) a. ***yerê*** ***hêłê***=*ş* *ard-**ê***
three egg.PL.DIR=3SG:A take.PST-**3PL:O**
'She took three eggs.' [HS.82]

b. ***penc*** ***çemçê***=*şa* *nîya=**n**=re*
five spoon.PL.DIR=3PL:A put.PST.PTCP.**M=COP.3SG.M:O**=POVB
'They (my family) had set [the cloth with] five spoons.' [JE.46]

This alternation in indexing in (2) is treated under differential object indexing (DOI). It is independent of the grammatical role but dependent on the inherent properties of the referent (e.g. animacy) or factors non-inherent to the referent, e.g. textual givenness. Put simply, not all object arguments are indexed on the verb (see Witzlack-Makarevich and Seržant 2020 for discussion and terminology).

The paper takes a corpus-based approach and quantitatively study object indexing in Hewramî. It aims to address the following questions: (i) the token

1 Throughout the paper, each example is being given a citation number, containing the initials of the narrative and the sentence number, indicating where the broader context for that example can be found in the text corpora (Mohammadirad 2025b, in prep).

frequency of object agreement in discourse and how the presence or absence of agreement relates to the overt versus zero realisation of object arguments; (ii) factors conditioning differential object indexing (DOI) in Hewramî in the presence of an overt O argument. The second question is further expanded into different sub-questions in §6. This is the first study of its kind to address a corpus-based account of differential object indexing in an Iranic language.

Some terminological conventions are in order. Following Haspelmath (2013), I use the term 'indexing' as a cover term for the 'agreement' phenomenon encompassing both 'grammatical agreement' and 'anaphoric agreement'. Indexing is a more neutral term than agreement and does not presuppose any syntactic dependency between the index and the co-referent nominal. The term 'agreement' is reserved for cases where the index obligatorily marks an argument regardless of the presence or the absence of the co-referent NP in the clause. Agreement in this sense parallels what Haig (2018) refers to as 'obligatory' indexing. On the other hand, the term 'alternating indexing' is used for cases where the presence of an index is conditioned by contextual factors, e.g. the absence of the co-referent NP. I use the following abbreviations to refer to the core arguments within the clause: S: single argument of an intransitive verb; A: agent of a transitive verb; O: patient of a transitive verb.

The paper is structured as follows. §2 looks into the theoretical concepts behind the complementarity hypothesis and DOI. §3 gives an overview of object indexing is past tense constructions in Iranic. §4 gives a brief overview of Hewramî and its typological profile. §5 looks into the token frequency of object indexing in Hewramî in light of the complementarity hypothesis. §6.1 introduces the corpus annotation behind the study and investigates the effect of individual variables behind DOI. The paper goes on to offer a multivariate analysis for the lack of otherwise present object index, predicting the probability of object indexing in Hewramî (§6.4). Section 7 is the discussion and conclusions.

2 Theoretical preliminaries

Agreement is commonly assumed to track reference and maintain salience in discourse (Barlow 1992). In light of this, the complementarity hypothesis argues that zero arguments are favoured by agreement markers. On the other hand, overt arguments are not favoured by agreement markers as such (Taraldsen 1978, Nichols 2018). According to Nichols (2018: 857) 'if agreement on verbs helps identify the referents of arguments, then agreement should be favored where arguments are likely to be non-overt (and vice versa)'. Thus, the first question aims to highlight the relationship between the presence of the agreement marker and the presence or absence of the co-referent O nominal in the same local domain. It is often assumed that the presence of agreement morphology in a language correlates with extensive pro-drop in discourse in that language. In the generative tradition of research on pro-drop, this is enshrined in the view that rich agreement 'licenses' pro-drop. The typological evidence for this is somewhat mixed: Nichols (2018) reports little evidence of complementarity in discourse from Nakh-Daghestanian languages, while Berdicevskis et al. (2020) support the claim for a trade-off between agreement morphology and overt pronouns, based on Slavic data and a larger typological sample.

Concerning the second question, factors conditioning object indexing may vary across languages. This has been treated under the umbrella term 'differential object indexing (DOI)' in recent scholarship (see Iemmolo 2011, Haig 2018, among many others). Put roughly, it means that not all object arguments are indexed in the discourse. This could be due to the inherent properties of the referent (e.g. animacy, humanness) or contextual factors (information structure, word order).

Differential indexing has been traditionally associated with topicality, in a way that topical referents tend to be indexed, whereas non-topical referents lack indexing (Givon 1976; Iemmolo 2011). The topicality of a referent is generally assumed to be enhanced by givenness, a high degree of identifiability and a high ranking in the person hierarchy. However, topicality is an umbrella term involving a great deal of vagueness. It has been used for a variety of different pragmatic effects across languages. Indeed, highly topical elements may be focused and, as a result, not indexed (see Just 2023 for a critical analysis).

Recent cross-linguistic studies have shown that there is no 'hard and fast rule' determining when the object arguments are indexed. Rather, differential object indexing results from a complex interplay of different factors. Following Haig (2018: 789), Just and Witzlack-Makarevich (2022) and Just and Čéplö (2022), the following factors have been listed as predictors of DOI.

- The presence/absence of an overt co-referential direct object in the local clause domain
- Word order
- PoS of the head
- Modification of the head
- Semantics of O (human, animal, object, etc.)
- Identifiability (definite, specific, non-specific)
- Textual givenness (given, new)
- The presence of an additional object that is higher-ranked in terms of animacy
- Person/number of the referent

In Maltese, following word order, definiteness is the most important factor triggering DOI (Just and Čéplö 2022). In Ruuli (Bantu, JE103), referential givenness is the most crucial factor, excluding word order as a predictor (Just and Witzlack-Makarevich 2022). The assumption is that differential indexing has language-specific, compositional causes and effects. In this paper, I take a gradient corpus-based approach to studying object indexing in Hewramî, mainly using the parameters just mentioned.

3 Variation in O indexing in Iranic languages

To better understand the variation in object indexing in the past tense within Iranic, it is useful to give a brief overview of the developments since Late Old Iranic period. A significant development in Iranic languages was the rise of ergativity by the early Middle Iranic period. The rise of ergativity was a by-product of changes to the nominal and verbal morphology (Haig 2008, 2017). The nominal morphology saw a reduction of an up to eight-term case system in Old Iranic to a two-term case system in Middle Iranic, namely the unmarked direct (or nominative) case and the oblique (non-nominative) case. As for the verbal morphology, the finite past tense forms were lost. As a result, the periphrastic perfect became the sole way of expressing past tense verb forms. In terms of argument structure, then, the resultative participle agreed with the direct-marked O-past argument as well as the S of intransitive constructions through a set of verbal person affixes attached to the stem of the verb 'to be'. On the other hand, the expression of A was carried out with a paradigm of pronominal clitics, see 3.1.

(3) *u=t* ***az*** *hišt* *h-**ēm*** *sēwag*
and=2SG:A 1SG.DIR left be-1SG:S orphan
'And you left me behind as an orphan.'
(Durkin-Meisterernst 2014: 394, paT.873) (Parthian)

Object indexing in past transitive constructions has seen many changes since the Middle Iranic period. In certain modern Iranic languages, e.g. Kurmanji, Hewramî, Zazaki, etc., it has been retained on the verb and may co-occur with a co-nominal, with the difference that the copula stem is now univerbated with the participle.

(4) *wî* ***ez*** *anî-**m***
3SG.OBL 1SG.DIR bring.PST-1SG:O
'He brought me [here].' (Northern Kurdish)

However, in most Iranic languages, the O index has degrammaticalised into a bound pronoun (or pro-index), meaning that it is no longer compatible with a co-referential nominal in the same local domain, see 5-6. This shift from an original agreement marker to a bound pronoun might be called ‘deinflectionalization’ following Norde (2009).

(5) *dena* ***ewan***$_i$*=î* *de-kuşt/*-****in***$_i$
otherwise them:O=3SG:A IPFV-kill.PST-3PL:O
‘Otherwise he would kill them.’
(Haig 2018: 803) (Central Kurdish)

(6) *aton* *ke* ***men***$_i$*=et* *sir* *va-kerd/*-****on***$_i$
Now that me:O=2SG:A well_fed PVB-do.PST-1SG:O
‘Now that you fed me.’
(Mohammadirad 2020: 417) (Delijani)

A further stage of decay in the ergative coding of the O argument is that in some modern languages, object indexing in the past tense has been levelled to the pattern in present tense constructions, meaning that pronominal clitics now alternatively index an object argument of a past tense clause as well. In other words, pronominal clitics have superseded grammaticalised inflectional person/number affixes in expressing past tense objects.

(7) *gorg* *eş=xa=****şû***
Wolf 3SG:A=eat.PST=3PL:O
‘The wolf ate them.’
(Mohammadirad 2020: 507) (Nowdani)

(8) *nîya=****şan****=im* *yane* *dayk=im*
leave=3PL:O=1SG:A house mother=1SG:PSR
‘I left them at my mother’s house.’
(Mohammadirad 2020: 376) (Gorani Qela)

In short, object indexing in past transitive constructions has developed radically since Old Iranic period. The original object agreement has degrammaticalised in most languages whereas S-past indexing is obligatory across the board, and A-past indexing is fully obligatory in most languages (see Mohammadirad and Haig forthcoming for a recent discussion of subject indexing in Iranian). The developments confirm typological evidence in object agreement being much less frequent than S/A agreement. Kibrik (2011) mentions the privileged role of ‘principal’ (=S/A) in agreement, and Bickel et al. (2013) report a strong bias towards S=A in agreement systems in the majority of genetic groupings.

Hewramî has preserved the split ergative construction in terms of the morphological coding: Inflectional person/number suffixes code S and O arguments in the past tense (see §4). What is essential for our discussion is that O-indexing inflectional affixes have remained nearly obligatory in Hewramî (see §5 for the token frequency of object agreement).

4 Language background

Hewramî is a member of the Gorani subbranch of the Iranic languages (Indo-European: Iranic: Central Iranic: Northwestern Iranic: (Adharic:) Gorani: Hewramî). It is spoken in a high mountainous region at the heart of the Kurdish-speaking region along the western border of Iran and neighbouring areas in Iraqi Kurdistan. Hewramî varieties are traditionally divided into three major groupings: Tekht, Luhon and Jawero. Geographically speaking, these varieties are spoken in the centre, south and east of the greater Hawraman region, respectively. MacKenzie (1966) is a sketch grammar of the Luhon variety. Mohammadirad (2026) is a comprehensive grammatical description of the Tekht variety.

Hewramî has a basic AOV word order. Like most Iranic languages, it has retained split ergativity: the alignment pattern is nominative-accusative in clauses derived from the present tense verbs and ergative-absolutive in clauses derived from the past tense verbs. Morphological ergativity is manifested both in terms of argument indexing and argument flagging (through case morphology, though not available for speech act participants). In terms of argument indexing, S and O are expressed by verbal person/number suffixes, while A is expressed by the clitic pronouns (column 3), see the paradigms of bound person marking in Table 1.

In terms of case marking, ergativity is manifested by morphological case marking on nouns and 3rd person pronouns. Hewramî has a two-term case system: ‘direct’ and ‘oblique’. Direct and oblique are Iranic internal terms, which in Hewramî are roughly equivalent to ‘nominative’ and ‘non-nominative’, respectively. The examples in (9) illustrate the ergative alignment in the past tenses. Note that the perfect

Table 1: Bound argument indexing in Hewramî (Mohammadirad 2026).

	S/A-present	S/O-past	A-past/O-prs
1SG	*-û*	*-a(nê)*	*=im*
2SG	*-î*	*-î*	*=it*
3SG	*-o*	*-Ø* (M); -e (F)	*=iş*
1PL	*-mê*	*-îmê*	*=ma*
2PL	*-dê*	*-îdê*	*=ta*
3PL	*-a*	*-ê*	*=şa*

verb forms in (9a) and (9b) are the same as the past forms, except they contain the copula stem *n*.

(9) a. ***jenî**=m* *merdê=**ne***
wife.F.SG.DIR=1SG:PSR die.PST.PTCP.F=COP.3SG.F:S
'My **wife** (S) died.' [ÇK.26]

b. ***jenî**=m* *ardê=**ne***
wife.F.SG.DIR=1SG:A take.PST.PTCP.F=COP.3SG.F:O
'I took a **wife** (O).' [JM.20]

c. ***jenê**=ç* *wat=**iş***
woman.F.SG.OBL=ADD say.PST=3SG:A
'The **woman** (A) said.' [HS.77]

Hewramî also features differential subject indexing (DAI) in past transitive constructions, where the expected A-indexing clitic may be absent due to issues associated with the A argument being focused (10)–(11), with a low effect of clause type and animacy (see Mohammadirad and Haig forthcoming for a corpus-based study). As seen later, DOI is rather triggered by factors different from DAI, e.g. person of the referent. The main property that seems to bring the two phenomena together, at least partially, is that the argument index tends to be more absent with overt inanimate subjects and objects.

(10) *çi* *melûm=a* ***to*** *koştê=b-a*
Ptcl obvious=COP.3SG.M:S 2SG kill.KILL.PST.PTCP.PL=be.PRS-3PL:O
'How can it be determined that **YOU** killed them?' [ME.153]

(11) *kuř-û* *min* ***xuđa-y*** *kuşte=n*
son-GEN.EZ 1SG God-M.SG.OBL kill.PST.PTCP.M=COP.3SG.M:O
'**GOD** killed my son.' [HM.95]

(12) *pejare-y* *berd-ø=we*
grief-M.SG.OBL take.PST-3SG:O=COMPL
'The grief overtook him.' [DB.245]

Drawing on two spoken corpora from the Tekht variety of Hewramî, this study gives the first detailed description of DOI in an Iranic language. The first corpus is a collection of 15 fully glossed narratives (Mohammadirad 2025b) and contains approximately 10,000 words. The second corpus is a collection of oral folktales collected by the author in Hawraman in August 2023. The corpus features 31 narratives and comprises around 25,000 words (Mohammadirad in prep). Overall, seven narrators (including two females and five males all over 50) were recorded telling folktales, local anecdotes and life stories (see Mohammadirad 2026 for details).

The corpus data were initially annotated in Excel. The annotated corpus is publicly available in a CSV file in the Zenodo repository (Mohammadirad 2025a). The subsequent investigation of indexing in Hewramî is based on the statistical analysis of these corpora.

5 O indexing Hewramî: an overview

As stated, Hewramî features split ergativity. In terms of O indexing, clitic pronouns (column 3 in Table 1) index objects. However, in this function, they remain alternating to the presence of the co-referent nominal. Thus, while verbal affixes indexing O-past arguments tend to be obligatory, the clitic pronouns indexing O-present arguments have remained pronominal, not only in Hewramî but across West Iranic (see Haig and Forker 2018; Mohammadirad 2020: chap.4).[2] This is illustrated by the examples in (13). In (13a), the independent pronoun *to* is the sole way to express the O. In (13b), the =*t* is in complementary distribution with the co-referent independent pronoun *to*.

(13) a.

to=*îç*	*ber-a*	*yane*
2SG:O=ADD	take.PRS.IND-3PL:A	house.M.SG.DIR

'They will take you to [their] house.' [HB.38]

b.

*ker-û=**t***	*pen-û*	*ʕałem-î*
do.PRS.IND-1SG:A=2SG:O	advice.M-GEN.EZ	world-M.SG.OBL

'I (hereby) expose your impropriety to the world. [Lit. I (hereby) make you the news of the world.]" [JP.198]

With regard to O-past indexing via suffixal morphology, the index may be present on the verb regardless of the presence or absence of the co-referent nominal. The following examples illustrate the compatibility of O indexing and the corresponding co-nominal.

(14)

to	***min***=*it*	*quł*	*kerđ-**a**=w*
2SG	1SG=2SG:A	pierced	do.PST-1SG:O=and

'You disabled me.' [PW.30]

2 Discussing the grammaticalization of object pronouns in Iranic, Haig (2018) holds that the grammaticalisation of object pronouns in Iranic is contrary to what is expected for subject agreement cross-linguistically: subject indexing is assumed to be derived from a cline of grammaticalization where independent pronouns gradually coalesce with the verbal host and turn into agreement markers. On the other hand, while object pronouns also lose prosodic independence early in the languages, they fail to develop into agreement markers and do not develop further than the differential O indexing state.

(15) *mar-êw-î* *gest-**a***
snake-INDF-M.SG.OBL bite.PST-1SG:O
'A snake bit me.' [MP.09]

The following examples illustrate object indexing in larger stretches of discourse. In both (16) and (17), the agreement markers first appear with the overt object NP in the same clause and then resume the absent O argument.

(16) *çage* ***to***$_i$*=şa* *sîyaw-e* *kerđ-**î***$_i$ *şîrîn-e* *kerd-**î***$_i$
There you:O=3PL:A black-F do.PST-**2SG:O** sweet-F do.PST-**2SG:O**
'There where they blackened you, [or rather] they sweetened you…' [XX.188]

(17) *be* *kune* ***awî***$_i$*=şa* *ardê=**ne***$_i$*=û*
By clay_pot water.F.SG.DIR=3PL:A bring.PST.PTCP.F=**COP.3SG.F:O**=and
*nîyɛ=**ne***$_i$*=şa=re*
put.PST.PTCP.F=**COP.3SG.F:O**=3PL:A=POVB
'They fetched$_i$ water$_i$ in clay pots. They unloaded it$_i$ [the water].' [JE.17]

This means that the index is present on the verb whether or not the co-referent NP is present in the clause. Yet, the token count of the object indexing in the corpus shows that there are more agreeing verbs with zero objects than with the overt objects (see Table 2).

As the data in Table 2 show, there is evidence for the complementarity hypothesis: zero objects are more likely to be indexed (90 %) than overt objects (82 %). This difference is also statistically significant. A Fisher's exact test gives a p-value of 0.0003, confirming a significant association between object type (overt vs. zero) and object indexing, indicating that object type influences the likelihood of indexing in past transitive clauses.

In corpus-based approaches, complementarity is understood as a gradient phenomenon: if there is a statistically significant association between, for example, the presence of a pronoun and the absence of an agreement marker, this is taken as evidence of some degree of mutual influence. As shown in Table 2, zero objects are associated with more indexing than overt objects, and this difference is statistically significant.

Table 2: Indexing O-past arguments.

	n. past tr. clauses	O is indexed		O is not indexed	
		N	%	N	%
Overt object	724	594	0.82	130	0.18
Zero object	606	545	0.90	61	0.10
Total	1330	1139	0.86	191	0.14

On the other hand, there are fewer agreeing verbs with overt objects (18 %) than with zero objects (10 %). These cases of lack of indexing constitute differential indexing of the O argument in Hewramî. The following examples are illustrative of the lack of indexing with overt plural object NPs.

(18) ***penc*** *çemç(e)-ê=şa* *nîya=**n**=re*
Five spoon.M-PL.DIR=3PL:A put.PST.PTCP.**M=COP.3SG.M:O**=POVB
'They (my family) had set [the cloth with] five spoons.' [JE.46]

(19) ***hîtêwe*** ***gaw****=iş* *sana-Ø*
a_pair cow=3SG:A buy.PST-3SG.M
'He bought a pair of cows.' [ED.29]

Finally, note that cross-linguistically, in languages with differential object indexing, the presence of indexing is the more unexpected pattern, and it is the restricted one (see Just 2024 for a recent overview). By contrast, in Hewramî, as seen in Table 2, the presence of the index is the default pattern. Viewed in this sense, lack of indexing constitutes the differentiality in Hewramî.

6 DOI in Hewramî

As seen in §5, the majority of objects are indexed, leaving the lack of indexing as the marked pattern in Hewramî. This section gives a detailed overview of DOI in Hewramî. Following an introduction into the corpus annotation and relevant variables (§6.1), we first look at the effect of individual variables on the DOI and then offer a multivariate analysis, delving into how different factors come together to shape the DOI. Recall that it is the lack of indexing that shapes the DOI in Hewramî.

6.1 Corpus annotation and relevant variables

This section lays out different factors behind DOI in Hewramî. Following the methodology used in Haig 2018: 789, Just and Witzlack-Makarevich 2022 and Just and Čéplö (2022), the corpus was annotated for the variables in Table 3. The variables feature three major properties. The first group containing slot co-optation (i.e. co-optation of the object index slot by another argument), constituent order and TAM refers to clausal properties. The second group containing PoS and NP modification refers to the syntactic properties of the object NP. The third group features animacy, identifiability, textual givenness and person of the referent, which refer to semantic-

Table 3: Variables.

Variable	Values	
Dependent variable	**Indexing**	**Index**, **No index**
	PoS of head	Noun, pronoun
	NP modification	Modified, non-modified
	Animacy	Human, (non-human) animate, inanimate
	Identifiability	Definite, specific, non-specific
Independent variables	Textual givenness	Given, new
	Slot co-optation	Yes, no
	TAM	Perfect, perfective, pluperfect, conditional past
	Person of referent	3SG.M, 3SG.F, 3PL, 1SG, 2SG, 1PL, 2PL
	Constituent order	OV, AOV, OVA, VO, OAV

referential properties of the object NP. The factors animacy, identifiability, textual givenness and person spell out topicality.

In surveying these different variables, I aim to answer the following questions. These questions are an expanded version of the second research question posed in the introduction section.

i. What factors license DOI in the presence of a co-referential NP in Hewramî?
ii. Which of the factors combined predict DOI? i.e. what factor makes it more probable that the object will not be indexed?
iii. What is the relative importance of different predictors in triggering DOI?

Question (i) seeks to answer which factors are individually relevant for DOI. For example, when discussing animacy, what is the effect of humanness on DOI? Question (ii) concerns the interplay of different factors in shaping DOI. For instance, how statistically significant is the person of the referent if another factor, such as givenness, is involved? Question (iii) attempts to identify the relative importance of factors in predicting DOI in Hewramî. For instance, how significant is the person of the O relative to givenness of the O in predicting DOI? As stated in §2, the variables have different weights in triggering DOI, and their interaction is subject to language-specific matters. These three research questions are interconnected and develop sequentially.

6.2 Individual variables

In this section, I investigate the weight of individual factors triggering DOI in Hewramî using descriptive statistics. In doing so, only factors that make statistically important contributions are discussed.

6.2.1 Slot co-optation

In a normal past transitive clause, an overt object is normally indexed on the verb, as shown by (20).

(20)	***duwanze***	***hêłê***=*ş*	*ard-**ê***
	Twelve	egg.PL.DIR=3SG:A	take.PST-**3PL:O**
	'She took twelve eggs.'		[HR.44]

However, when a higher-ranked argument, such as the recipient, beneficiary, addressee, etc., is indexed on the verb, this comes at the cost of expressing object indexing. Following Haig (2018), I refer to this phenomenon as 'slot co-optation', meaning that the object agreement slot is co-optated by another argument. In (21), one would expect the 3PL object to be indexed on the verb, as in (20) above. However, the agreement slot is co-opted by the higher-ranked recipient argument, which is pronominally indexed on the verb. In (22), the verb does not index the 3SG.F direct object, expressing instead the beneficiary argument.

(21)	*sênze*	*danê*	*heser(e)-ê=şa*	*da-**ø***	***pene***
	thirteen	CLF.PL	mule.F-PL.DIR=3PL:A	give.PST-3SG:R	to
	'They gave him thirteen mules.'				[ÇH.69]

(22)	*sêłê=şa*	***pey***	*kerd-**ø***
	halva.F=3PL:A	for	do.3SG.M:R
	'They made halva for him.'		[MM.35]

This phenomenon is common in some Iranic languages, e.g. Laki, Central Kurdish (Öpengin and Mohammadirad 2022), and has parallels cross-linguistically, e.g. in Warlpiri (Hale 1982: 251–252). In Hewramî, the corpus data show that DOI is categorically associated with slot co-optation, meaning that the object index is absent in the presence of indexing for a higher-ranked argument on the verb. This is evidenced by the fact that none of the 54 examples from our corpora that feature the indexing of a non-core argument index a direct object NP. Put differently, if the O-index slot is filled by the non-core argument, the object cannot be indexed in Hewramî. Overall, slot co-optation makes up 41.5 % (54 out of 130 tokens) of cases where object indexing is absent in the presence of a co-referential object, making it the most important predictor triggering DOI; see §6.4.

A note on the statistical description of the rest of the variables is in order. Since slot co-optation is a syntactic rule categorically associated with DOI, it was left out in discussing other individual variables. Thus, when discussing the effect of person, all cases where slot co-optation has been coded as "yes" are cast aside. This guarantees a fair analysis of the impact of individual variables.

6.2.2 Person of the referent

The Agreement relationship may be sensitive to the person of the referent. Speech act pronouns, as the most accessible referents in discourse, tend to trigger agreement on the verb more than other persons (Ariel 2000). This is also reflected in our data. Out of 20 tokens with the direct object having SAP referents, only in one token does the verb not index the person of the object. The rate of non-indexing is 6.5 % for direct objects with 3SG.F referents (8 out of 122) and highest for 3PL referents, with 44 % not being indexed (67 out of 153 tokens), see Figure 1. The difference in non-indexing between 3PL and other persons (including 3sg.m: 375 indexed; 0 no index) is also statistically significant (Fisher's exact test P-value equals $p < 0.0001$).

The following examples illustrate the lack of indexing with direct objects with different referents.

(23) ***min**=îç=şa* *ça* *sîyaw-Ø* *kerđe-**n***
1SG=ADD=3PL:A there black-M do.PST.PTCP.**M=COP.3SG:O**
'They blackened me (m.) too there.' [XX.188]

(24) ***awî**=ş* *kuşte=**n***
water.F.SG.DIR=3SG:A stop.PST.PTCP.**M=COP.3SG.M:O**
'He stopped the water [flow].' [MF.309]

(25) ***girđ -û*** ***ijdeha-y-eka*** *melekuř-a* *ward*
all-GEN.EZ serpent-EP-DEF.PL.OBL grasshopper-PL.OBL eat.PST
'The grasshoppers ate all the serpents.' [ŞŞ.14]

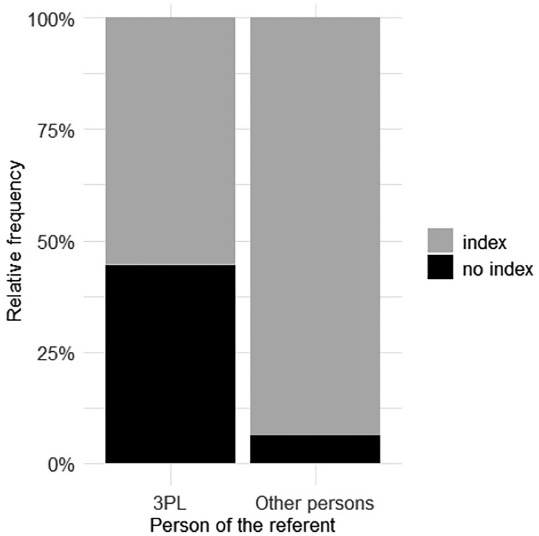

Figure 1: Object indexing and person of the referent.

The higher rate of DOI for 3PL referents may suggest that there is some semantic basis to object agreement in Hewramî. Here, conjoined singular noun phrases were also counted as 3PL. With these kinds of referents, there is evidence that animacy may play a role in whether or not the verb agrees with the direct object. In (26), the verb does not agree with the inanimate conjoined NPs, occurring instead in its bare 3SG.M form. However, in (27), the verb agrees with conjoined NPs that are human.

(26)	***liç*=*û***	***lût-û***	***dêw-eka***=*ş*	*berđ* **-Ø**
	lip=and	nose-GEN.EZ	ogre-DEF.PL.OBL=3SG:A	take.PST-3SG.M:O

‘He took the ogres’ **lips and noses** [to the king].’ [ME.156]

(27)	***Î***	***jen(î)-û***	***wê*=*m*=*e*=*w***	***î***
	DEM.PROX	wife-GEN.EZ	REFL=1SG:PSR=DEM=and	DEM.PROX
	kuř*=*m*=*e	*kuşt***ê=bî-ɛn-ê**	*êtir îse*	
	son=1SG:A=DEM	kill.PST.PTCP**.PL=be.PST-COND.AUG-3PL:O**	well now	
	min	*çêş*=*im*	*kerd-ɛ*	
	1SG	what=1SG:A	do.PST-COND.AUG	

‘Had I killed my wife and my son, what would I have done now?’ [XŞ.104]

Table 4 summarises the percentage of object indexing according to whether the conjoined phrases are animate (in this case, human) or not. Even though there are only two tokens with conjoined noun phrases that are animate, tentatively, it can be said that there is a sharp difference in animacy between conjoined noun phrases and whether or not they trigger agreement on the verb. The high indexing tendency with conjoined NPs that express animate referents further confirms the typological tendencies laid out in Corbett (2006: 184–185).

In conjoined noun phrases denoting inanimate entities, when verbal agreement is controlled by only one conjunct, the second conjunct generally triggers agreement. In (28), ‘spleen’ and ‘liver’ have different genders; the verb agrees with the nearest conjunct.

Table 4: Indexing and the conjoined phrases.

Types of conjoined NPs	n. past tr. clauses	O is indexed	
		N	%
Animate	2	2	100 %
Inanimate	27	2	7 %

(28) *yo=şa* *şimş(î)=û* *yeher-eke=ş*
one3PL:PSR spleen.F=and liver.M-DEF.M.SG.DIR=3SG:A
warde=n
eat.PST.PTCP.M=COP.3SG.M:O
'One of them ate the [dove's] spleen and liver.' [DB.83]

6.2.3 NP modification

Under NP modification, each of the object NPs was coded for whether they were bare or modified. By modified, we mean whether numerals, quantifiers and demonstratives modify the head noun. Additionally, conjoined noun phrases were counted as modified. The following examples illustrate DOI with NP modification.

(29) *gir* *ker-ê=şa* *kerd-Ø*
All chore-PL.DIR=3PL:A do.PST-3SG.M
'They did all [their] chores.' [HB. 58]

Note that numerals generally trigger plural inflection on direct object NPs, as seen in (2a)-(2b). Some head nouns do not carry number agreement that is triggered by numeral and measure nouns. In such cases, the verb tends to agree with the morphologically singular head noun; (30)-.

(30) *wîs* *kołê* *loke=şa* *ard-Ø*
Twenty load.PL.DIR cotton=3PL:A bring.PST-3SG.M
'They brought twenty loads of cotton.' [ME.120]

(31) *ewt* *ko(e)-ê* *xezêne=ş* *berđ -Ø=we* *la-w*
Seven load-PL.DIR treasure=3SG:A take.PST-3SG.M:O=COPL to-EZ
mamojenî=ş
uncle's_wife=3SG:PSR
'He took back seven loads of treasure to his uncle's wife.' [ED.139]

Similarly, when the head noun modified by a numeral is marked by the definite suffix *-e*, the plural suffix is absent on the head noun. In such cases, the modified NP may still trigger plural agreement on the verb (see 32). This could be regarded as semantic agreement as the head noun doesn't have plural morphology, but is semantically plural, and this plurality in turn triggers agreement on the verb.

(32) *ta* *çil* *kuř-e=ş* *berđ –ê*
until forty son-DEF=3SG:A take.PST-3PL:O
'Until he took all the forty sons.' [ÇH.08]

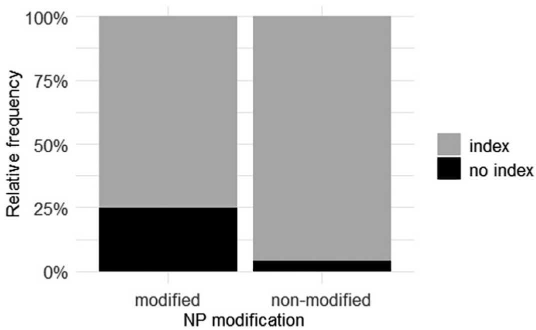

Figure 2: Object indexing and NP modification.

Overall, 25 % (58 out of 231) of modified Os are unindexed, while 4 % (18 out of 439) of non-modified Os are unidexed. This is visualised in Figure 2. Modified NPs are associated with less indexing than bare ones. The difference in unindexing is statistically significant as well (Fisher's exact test gives p < 0.0001). The higher rate of indexing with bare NPs can be linked to the generally higher accessibility of bare nouns in discourse, compared to NPs containing full names and long definite descriptions (Ariel 2000). Similarly, Just (2023) suggests that referents that require modification can be considered less identifiable than referents that don't require modification.

6.2.4 Animacy

It has been suggested that referents expressing a high degree of animacy, e.g. human, are associated with more topicality and accordingly being kept track in discourse through indexing (Iemmolo 2011). The following examples illustrate DOI with inanimate (33) and human (34) referents.

(33) ***çaşt(î)-ekê***=*şa* *kerde*=***n*** *Be*
meal.F-DEF.F.SG.DIR=3PL:A do.PST.PTCP.**M=COP.3SG.M:O** on
awîrga-(e)kê=û
hearth.F-DEF.F.SG=and
'They made the food on the hearth.' [JE.39]

(34) ***nicûmger-ekê***=*ş* *nîya-Ø*
astrologist-DEF.PL.DIR=3PL:A put.PST-3SG.M:O
'He (Pharoa) appointed the astrologists.' [MF.36]

However, when examined on corpus counts, it turns out that the rate of DOI (i.e. non-present indexing) is higher for inanimate direct objects than animate ones. Out of 429 inanimate Os, 15 % (63) are unindexed. The unindexing rate is 11 % (6 out of

54 tokens) for non-human animate referents and 4 % (7 out of 187) for human referents. This difference is in line with typological tendencies; there is more zeros (here lack of indexing) with inanimate referents (Haig and Schnell under review). However, this is not in line with accessibility theory, which predicts more attenuated forms of expressions (here zero) will be used with more animate referents in discourse.

When tested pairwise, not all differences reach significance, but this is partly a consequence of low absolute numbers in some of cells (Table 5). Notably, the difference in animacy is statistically significant between inanimate and human referents and between humans and non-humans (Figure 3).

Figure 5 visualises the difference in DOI between human and non-human referents.

6.2.5 Identifiability

Identifiability and givenness express the information-structural status of the referent, high degree of which is assumed to increase the topicality of referents, consequently leading to more indexing. As for identifiability, object NPs were

Table 5: Pairwise testing for significance of values along animacy lines.

Categories	Fisher exact value	p<0.05
Inanimate vs animate	0.679	No
Animate vs human	0.0788	No
Inanimate vs human	0.000028	Yes
Inanimate + animate vs human	0.000038	Yes

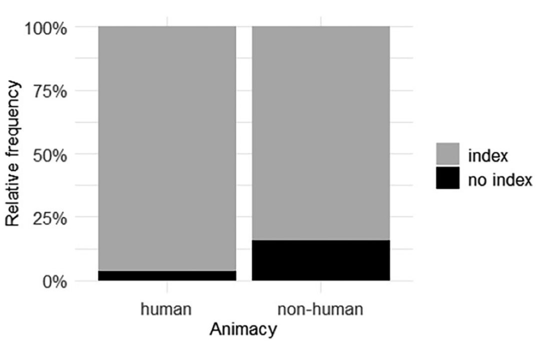

Figure 3: Object indexing and animacy.

annotated as definite, specific and non-specific. The following examples illustrate DOI with definite (35), specific (37) and non-specific referents (36).

(35) *çem-ê=ş* *řeşt*
eye-PL.DIR=3SG:PSR/A apply_Kohl_to_eyes
'He applied Kohl to her eyes.' [SH.167]

(36) *yanekołê=şa* *kerde=n*
moving_house.F=3PL:A do.PST.PTCP.M=COP.3SG.M:O
'They moved house.' [Lit. did house moving] [ZB.22]

(37) *beʕzê=ş* *piřna=ş=re*
some.PL.DIR=3SG:PSR break_off.PST=3SG:A=POVB
'It [the flood] took [and] threw some of them (the animals) [into the river].' [JE.71]

Out of 409 nouns with definite reference, 9 % (37) are unindexed. In contrast, the rate of unindexing is 16 % for direct objects with specific reference (25 out of 152) and 13 % with direct objects with non-specific reference (14 out of 109). Fisher's exact tests confirm that only pairwise testing for definite versus specific is significant ($p = 0.0155$). The pairwise testing for definite versus non-specific ($p = 0.276$) and specific versus non-specific ($p = 0.483$) doesn't reach significance. When specific and non-specific objects are combined, the difference between definite and non-definite objects is 0.0241, which is below 0.05 threshold and is thus significant.

6.2.6 Givenness

Givenness, like identifiability, expresses the information-structural status of the referent. It is expected that new referents are less indexed. The following examples illustrate DOI with referents, which are new (38), and given (39).

(38) *çêrxan=û* *serxan-êw=şa* *weş* *Kerđ*
basement=and upper_floor-INDF=3PL:A right do
'They constructed a basement and an upper floor.' [DB.37]

(39) *name-(e)kê* *wê=ş* *niwîs* *ser*
name.F.SG.DIR-DEF.F REFL=3SG:A write.PST on
maziłox(e)-eka=we
prayer_rug-DEF.PL.OBL=POST
'He wrote his name on the prayer rugs.' [ŞE.79]

Excluding cases with slot co-optation, the percentage of differentially indexed objects with new referents is 14 % (27 out of 187), against 10 % (49 out of 483) for given referents. However, the pairwise testing of differentially indexed objects (new referents vs given referents) is at a p-value 0.135, indicating no statistically significant association between givenness and differential object indexing.

6.2.7 Part of speech of the O NP

The next predictor is part of speech of the O NP. This variable has two values: noun versus pronoun. The latter includes personal, interrogative and indefinite pronouns. The following examples illustrate DOI with pronominal and nominal objects.

(40) *ewê=ş* *nîya* *bira-w* *wê=ş*
3PL.DIR=3SG:A put.PST.3SG.M brother.GEN.EZ REFLX=3SG:PSR
'He made them (the ogres) his brothers.' [ME.99]

(41) *gilke=ş* *kerđ =ne*
finger.F=3SG:PSR/A do.PST.3SG.M=POVB
'He put his finger [inside the hole of the mill].' [BM. 113]

The corpus counts show that of 556 nominal Os, 62 (i.e. 11%) lack indexing. Pronominal Os exhibit 12% (14 out of 114) differentiability. When tested pairwise based on Fisher's exact test, the difference between differentially indexed nouns and pronouns reaches a p-value of 0.746, indicating that PoS doesn't have a significant role in predicting DOI.

6.2.8 Constituent order

The next individual factor is constituent order. The observed orders in the database (excluding co-optation) are OV, AOV, OVA, VO and OAV. Table 6 summarises DOI per constituent order.

Table 6: The effect of constituent order on DOI.

	n. past tr. clauses	O is indexed		O is not indexed	
		N	%	N	%
OV	524	459	0.88	65	0.12
AOV	106	97	0.92.5	9	0.07.5
OAV	27	25	0.93	2	0.07
VO, OVA	13	13	0.100	0	0.00

The two major orders with relatively high number of DOI are AOV and OV. A Fisher's exact test gives a p-value of 0.321 indicating that the difference in unindexing between these two orders is not statistically significant. Similarly, there is no statistically significant difference between AOV and OAV (p= 1.00), suggesting that whether or not O is in the immediate preverbal position has no affect in its differential indexing. Note, however, that the tokens for OAV order are low, rending the results tentative. Crucially, the O is topical in OAV order, as it does not appear in the immediate pre-verbal slot. The verb in such cases predominantly agrees with a topical object (93 % of the cases). The following examples represent OAV order with (42) and without (43) object indexing. Note that in both examples A indexing is absent due to differential A indexing (see Mohammadirad and Haig forthcoming for a detailed discussion).

(42) *Î* *dijmen-ê=m=e* *kê*
DEM.PROX enemy-PL.DIR=1SG:PSR=DEM Who
kuştê=nê
kill.PST.PTCP.PL=COP.3PL:O
'Who killed my enemies?' [ME.150]

(43) *girđ -û* *ijdeha-yeka* *melekuř-a* *ward-Ø*
all-GEN.EZ serpent-DEF.PL.OBL grasshopper-PL.OBL eat.PST-3SG:O
'The grasshoppers ate all the serpents.' [ŞŞ.14]

6.2.9 TAM

Finally, we look at the effect of tense on differential O indexing. The relevant verbal TAM categories are perfective (or simple past), perfect, pluperfect and past conditional.[3] According to Table 7, DOI is almost entirely limited to perfective and perfect tenses. This is probably partly due to their higher token frequency compared to other TAMs. The pairwise difference between these two verbal categories gives a p-value of 0.290, indicating that the difference in unindexing between these two categories is not statistically significant.

6.3 Summary of individual variables

This section laid out the effect of individual variables on DOI. In short, slot cooptation always predicts DOI. The effect of person of the referent, NP modification, and animacy, and identifiability was seen to be statistically significant. The rest of

3 In Hewramî, past imperfect has nominative-accusative alignment and is thus excluded here.

Table 7: The effect of TAM on DOI.

	n. past tr. clauses	O is indexed		O is not indexed	
		N	%	N	%
Perfective	444	388	0.87	56	0.13
Perfect	198	179	0.90	19	0.10
Pluperfect	21	20	0.93	1	0.05
Conditional	7	7	0.100	0	0.00

factors including givenness, constituent order, TAM and PoS had little effect on DOI, confirmed by the results from Fisher's exact test.

While the simple statistics laid out here indicate the effect of individual values within each variable on the DOI, it cannot reveal how different factors come together in shaping the DOI. For example, the interplay between person, animacy and givenness cannot be captured. Secondly, simple statistics fail to capture the relative importance of these variables in impacting DOI. In the next section, these aspects will be addressed using multivariate analysis.

6.4 Predicting the probability of object indexing in Hewramî

This section examines which combination of factors predicts the probability of lack of indexing in Hewramî. The basic assumption is that DOI is the result of the combined effects of relevant variables/factors. For this reason, conditional inference trees (CITs) were used to illustrate the mentioned interplays. CITs test recursively how the presence or lack of verb indexing (dependent variable, index) is influenced by predictor variables, i.e. person, NP modification, animacy, etc. Unlike traditional decision trees (e.g. classification and regression trees), CITs use a significance testing procedure to select predictors at each split (with p-values obtained from permutation tests) and tend to reduce overfitting.

Note that I only examine the factors conditioning DOI in the presence of a coreferential NP. Slot co-optation has not been included in the model as it is categorically associated with the absence of O indexing (see §6.2.1). The reason is that its inclusion would interfere with further splitting in the model. Figure 4 illustrates a decision-making process for the presence (illustrated in grey colour) or lack of indexing (illustrated in black) where predictor variables are represented by nodes. Nodes, in turn, are split based on statistical significance and along the values of a

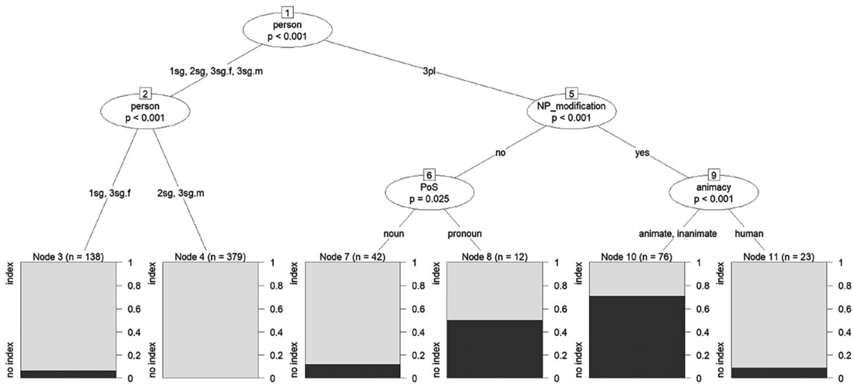

Figure 4: Predicting DOI without slot co-optation as a predictor.

predictor. I used the ctree() function from the party package in R (Hothorn et al. 2006) to construct conditional inference trees.

The model in Figure 4 shows that without slot co-optation as a potential factor, person of the O argument is the strongest predictor of DOI in Hewramî. The model suggests an initial split between 3pl and other persons. For the first group, NP modification (node 5) is the next strongest predictor. If the NP is not modified (node 6), then there is a weak effect of PoS where pronominal Os (node 7) show more DOI than nominal Os (node 8). If the NP is modified, animacy (node 9) plays a key role, with non-human referents (node 10) featuring more DOI than human referents (node 11). If the person of the referent is other than 3pl, there is further branching for the category person (node 2), consisting of other persons. Here, person has only a weak effect on DOI with 3sg.f and 1sg being associated with around 5 % of lack of indexing (node 3), though recall from §6.2.2 that there is only one token exhibiting lack of object indexing with an overt 1sg object. On the other hand, 2sg and 3sg.m referents always trigger agreement on the verb (node 9).

It is notable that identifiability was not predicted as important variables, despite the fact that it was reported to be statistically significant for definite versus non-definite. In other words, the significant differences observed for identifiability do not carry over in the multivariate analysis. Once person, NP modification and animacy are taken into account, identifiability become non-influential to predict DOI.

The prediction accuracy for the model in Figure 4 is 92.57 %, higher than the 82 % baseline for the majority class 'indexing'. A 95 % confidence interval for the model is between 88.05 % and 95.78 %. The statistical comparison of the model's accuracy (92.57 %) to the baseline (82 %) yields p = 0.02676. The precision of the model predicting the positive value (i.e. indexing) is 96.57 %. The recall (or sensitivity) is 94.94 %. The negative predictive value is 66.67 % (see Table 8 for other information).

Table 8: The evaluation of the conditional inference tree model.

Prediction accuracy	92.57 %
Balanced accuracy	84.97 %
Sensitivity (recall for 'index')	94.94 %
Specificity (recall for 'no index')	75 %
Positive predictive value (precision for 'index')	96.57 %
Negative predictive value (precision for 'no index')	66.67 %
Prevalence of 'index'	88.12 %
Detection rate (correct predictions of 'index')	83.66 %
Detection prevalence (predicted 'index')	86.63 %
McNemar's test p-value	0.2673

Looking now at the presence of the O index on the verb, it can be said that it is associated with referential and semantic features of the referent. The referential component concerns the person of the referent, with all persons other than 3pl strongly favouring indexing, and the semantic component concerns animacy. In the case of animacy, the value which is associated with more indexing, namely, human, is considered to enhance topicality. In the case of the person of the referent, the split is not straightforward, but SAP pronouns tend to trigger more indexing on the verb, in line with predictions of accessibility theory.

The conditional inference model duly predicts the complex interplay between different variables; however, as there is no single final tree generated by the model, the results may be hard to interpret. There are ways to deal with this issue, including using gradient boosting models, which can determine how often a particular predictor was selected for a 'split' across many thousand iterations of trees generated by the model. Figure 5 provides the answer to the question of the relative importance of factors affecting DOI.

Unsurprisingly, slot co-optation has been given predominant importance affecting DOI with a relative frequency of 43 %. Person is the next factor with high relative frequency of around 30 % affecting DOI. The high impact of person aligns with the first split in the conditional inference tree model. Animacy and NP modification contribute slightly to DOI, 9 % and 7 %, respectively. The rest of the factors, including TAM, identifiability, constituent order, PoS and givenness, have 3 % or less effect on DOI.

The accuracy of the gradient boosting model (or XGBoost) is 95.9 % (95 % CI: 91.3–98.5 %). The model reaches a Kappa of 83.3 %, a balanced accuracy of 90.9 % and a specificity of 98.36 %. The precision of the model predicting the positive value (i.e. indexing) is 94.77 %. The recall (or sensitivity) is 83.33 %. The positive predictive

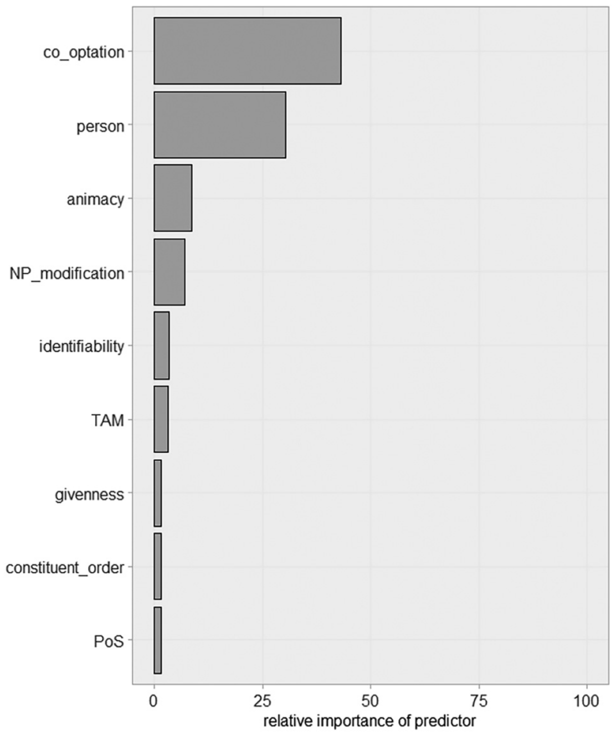

Figure 5: Relative importance of different factors affecting DOI.

value was 0.90.91 %, and the negative predictive value was 96.8 %, showing reliable classification for both categories.

Compared to the 92.57 % accuracy of CIT model, the boosting model shows slightly higher overall predictive performance (93.79 %). The CIT model exhibited higher sensitivity (94.94 % vs 79.31 %) but lower specificity (75.0 % vs 97.41 %) relative to XGBoost, indicating that CIT is better at identifying index (the positive class) but less accurate at correctly classifying no index.

7 Discussion

This paper offered a corpus analysis of object indexing in past transitive constructions of Hewramî. In doing so, I first examined the complementarity hypothesis against Hewramî data: in terms of token frequency, there were more indexing verbs with zero objects (90 %) compared to overt objects (82 %), providing thus some support for the complementarity hypothesis.

The paper then examined differential indexing of overt objects in Hewramî, where the absence of indexing constituted the differentiality. Following the relevant studies in the literature, the corpus was annotated for different variables, including slot co-optation, person, animacy, NP modification, identifiability, givenness, TAM, constituent order and PoS of the head. I then examined how the values within each variable may individually affect DOI. It was seen that DOI is categorically associated with slot co-optation, which means that if the O-index slot is filled by the non-core argument, the object cannot be indexed. In the category of person, DOI is most commonly in place when the person of the referent is 3PL. In terms of animacy, the impact of non-human referents on DOI is higher than human ones. There is also some effect of NP modification and identifiability. Other factors were seen to be statistically irrelevant for explaining DOI.

Gradient-boosting models were used to identify the interplay between different predictors and to identify the relative importance of different predictors in impacting the DOI. As for the first aspect, it was seen that for 3PL referents, NP modification comes together with animacy to affect DOI. As for the second aspect, it was seen that slot co-optation, person of the referent, and (less so) animacy and NP modification are relatively high predictors of DOI in Hewramî. In contrast, other predictors, including TAM, givenness, constituent order and PoS of the head, showed very little importance for predicting the absence of object indexing on the verb.

The findings of this paper could potentially inform the deconstruction of ergativity in Iranian languages, particularly in terms of argument indexing. As seen, the default object agreement completely disappears when its slot is taken over by indexing for a higher-ranked argument, usually a recipient, beneficiary or a possessor. The object agreement then starts to disappear with conjoined noun phrases expressing inanimate referents, and with non-human modified NPs expressing 3PL referents.

Research on differential object indexing ascribes the differentiality to a high degree of topicality and salience in discourse (e.g. Iemmolo 2011). However, as remarked, topicality is an umbrella term encompassing different semantic-referential properties and is enhanced by a high level of identifiability, animacy and givenness. Indeed, the Hewramî data exhibit a high level of indexing for the humanness that could potentially imply a greater degree of topicality.

In conclusion, it is the combined effect of animacy along with the person of the referent and NP modification factors that best spells out DOI in Hewramî. Overall, comparing the DOI in Hewramî with corpus-based studies on other languages reveals that while topicality may be a strong indicator of differential indexing, the conditions that trigger DOI are language-specific.

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Abbreviations

A	transitive subject
AUG	past convertor augment
CLF	classifier
COND	conditional
COP	copula
DEF	definite
DEM	demonstrative
DIR	direct case
GEN.EZ	genitive ezafe
IND	indicative
INDF	indefinite
IPFV	imperfective
M	masculine
O	transitive object
OBL	oblique case
PCPL	participle
PL	plural
POVB	postverb
PRF	perfect
PROX	proximal
PRS	present
PSR	possessor
PST	past
R	non-core argument
REFLX	reflexive base
S	intransitive subject
SG	singular

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